Experiment No. 20 – Dilation Morphological Operation

1. Aim

To implement the Dilation technique as a Morphological Operation to expand the foreground regions using OpenCV.

2. Program

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()
filename = list(uploaded.keys())[0]

# Read image
image = cv2.imdecode(
    np.frombuffer(uploaded[filename], np.uint8),
    cv2.IMREAD_COLOR
)

if image is None:
    print("Image loading failed")
else:
    # Convert to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

```
# Convert to binary image
_, binary = cv2.threshold(
    gray, 127, 255, cv2.THRESH_BINARY
)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Apply dilation
dilation = cv2.dilate(
    binary, kernel, iterations=1
)

# Display output
plt.figure(figsize=(12, 5))

plt.subplot(1, 3, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(binary, cmap="gray")
plt.title("Binary Image")
plt.axis("off")
```

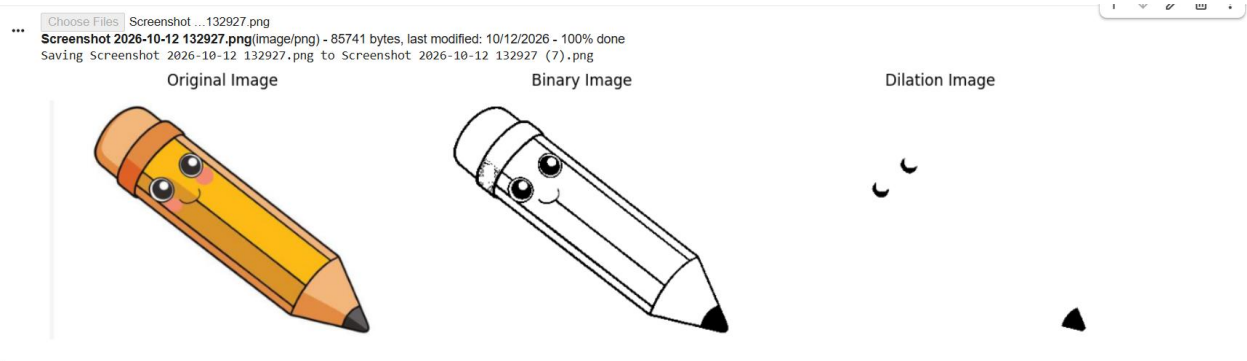
```
plt.subplot(1, 3, 3)
plt.imshow(dilation, cmap="gray")
plt.title("Dilation Image")
plt.axis("off")


plt.tight_layout()
plt.show()
```

3. Expected Output

The output displays the original image, binary image and dilated image. The white foreground regions become thicker and larger.

4. Output



The foreground regions are successfully expanded using the dilation operation.

5. Result

Thus, the Dilation Morphological Operation was successfully implemented using Python and OpenCV.